Ride-Hailing Operations Dashboard 2025

العربية

Overview

A multi-page Power BI report built on a **star schema** data model, with data preparation in **Power Query** and calculations in **DAX**.

Business Problem

A ride-hailing company wants to understand demand, cancellations, and revenue across cities and zones.

Questions

- Which cities, zones, and vehicle types drive the most trips and revenue?
- When are the peak hours?
- What is the cancellation rate, and when does it rise?
- How does surge pricing relate to cancellations and ratings?

Dataset

- **Synthetic dataset** generated for practice (not real trips)
- 6 tables in one Excel workbook:
  - `trips` (fact table): about 15,000 trips in 2025
  - `drivers` : 320 drivers
  - `riders` : 2,600 riders
  - `zones` : 27 zones in 5 cities
  - `vehicle_types` : 4 types
  - `payment_methods` : 5 methods

Tools

- Power BI

- Power Query
- DAX

Data Model

- Star schema: `trips` in the center, dimension tables around it
- One-to-many relationships from each dimension to `trips`
- A date table and an hours table, both created in Power Query
- `trips` connects to `zones` twice (pickup and dropoff):
  - Pickup relationship: **active**
  - Dropoff relationship: **inactive**, used in measures with `USERELATIONSHIP`

Data Preparation (Power Query)

Issue	Fix
Duplicate trips (25 rows)	Removed on `Trip_ID`
Date and time in one column	Split into `Date` (linked to the date table) and `Hour` (linked to the hours table), original column removed
Missing tips	Replaced with 0, because no tip means zero
Missing ratings	Kept as `NULL`, because no rating is not a rating of zero
Cancelled trips have fare 0	Kept, and fare averages are calculated on completed trips only

DAX Measures

- Total Trips, Completed Trips, Cancelled Trips
- Cancellation Rate
- Total Revenue, Avg Fare, Avg Distance
- Avg Rating, Avg Surge
- Dropoff Trips (uses `USERELATIONSHIP`)

Report Pages

1. **Overview:** trips, revenue, and payments at a glance
2. **Peak Hours and Cancellations:** when do riders cancel, and why?
3. **Zones:** where is demand highest?

Key Findings

- The overall cancellation rate is **12.08%**.
- [Add your finding on peak hours after checking page 2]
- [Add your finding on surge pricing, cancellations, and ratings]
- [Add your finding on the zones page]

Limitations

- The data is synthetic, so patterns reflect how the data was generated, not a real market.
- Only one year (2025) is included, so year-over-year comparison is not possible.

Files

- `ride_hailing_data.xlsx` : dataset (6 sheets)
- `ride_hailing_dashboard.pbix` : Power BI report
- `images/` : screenshots of the three pages and the model view

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